AI Logistics Control Tower & Intelligent Operations Platform

Total Orders

988

Count of Order_ID

Avg Time

6.52

Average of Delivery_Time

Avg Cost

821.80

Average of Shipping_C...

Avg Rating

3.05

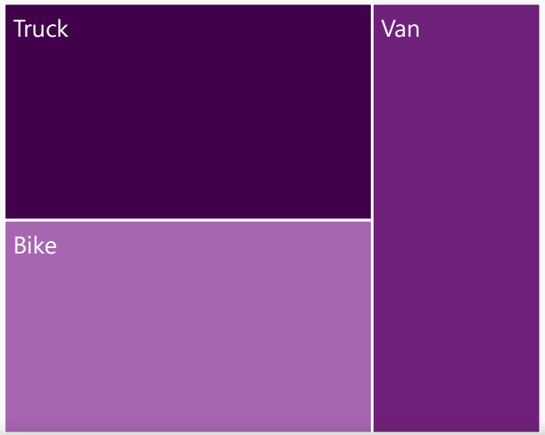
Average of Driver_Rating

Avg Dist

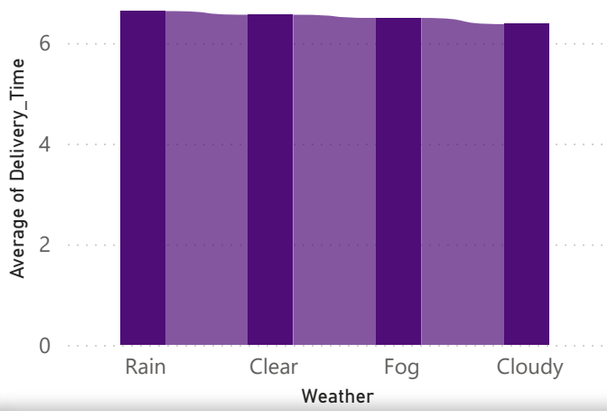
53.51

Average of Distance_km

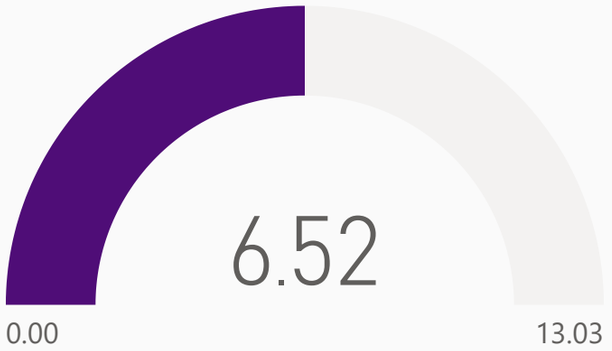
Orders by vehicle type



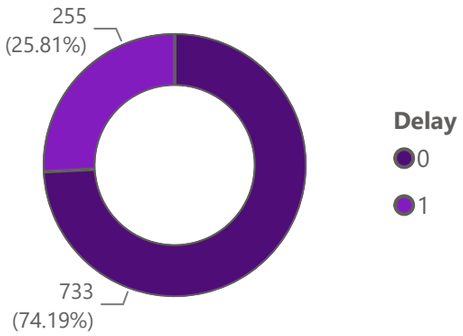
Average of Delivery Time by Weather



Average of Delivery Time



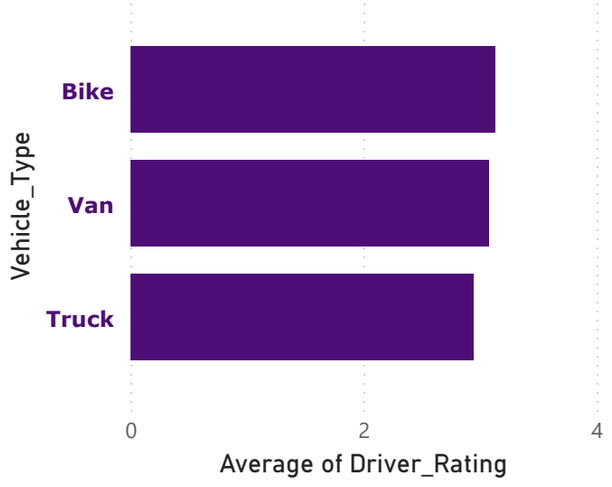
Count of Order_ID by Delay



Vehicle_Type

- Bike
- Truck
- Van

Average of Driver Rating by Vehicle Type

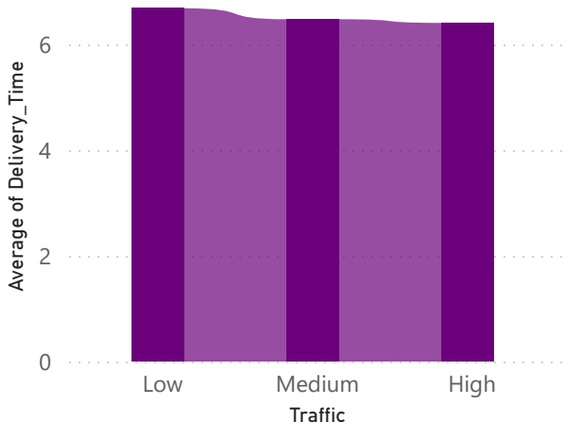


AI Logistics Control Tower & Intelligent Operations Platform

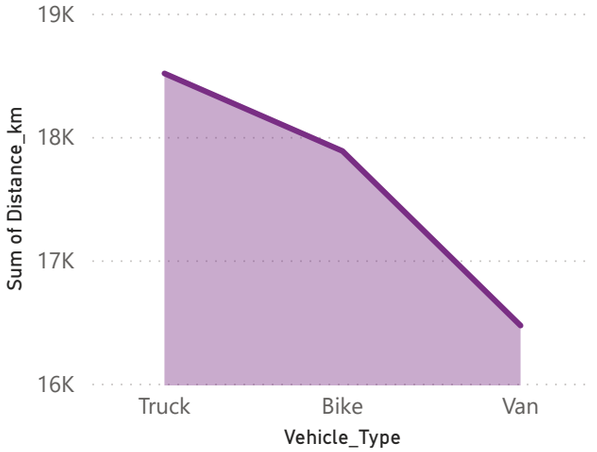
Average of Shipping_Cost by Weather



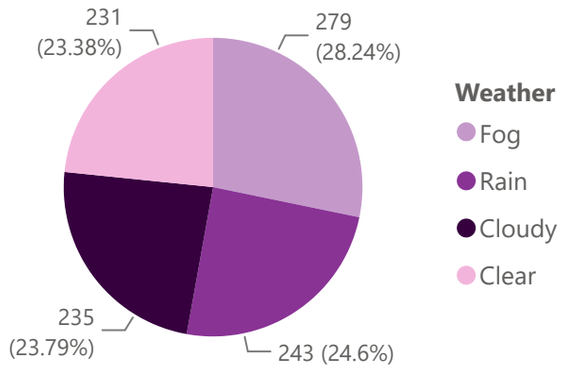
Average of Delivery_Time by Traffic



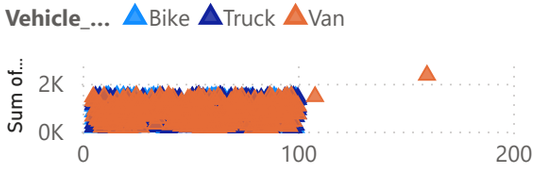
Sum of Distance_km by Vehicle_Type



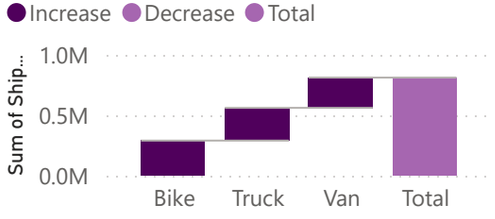
Count of Order_ID by Weather



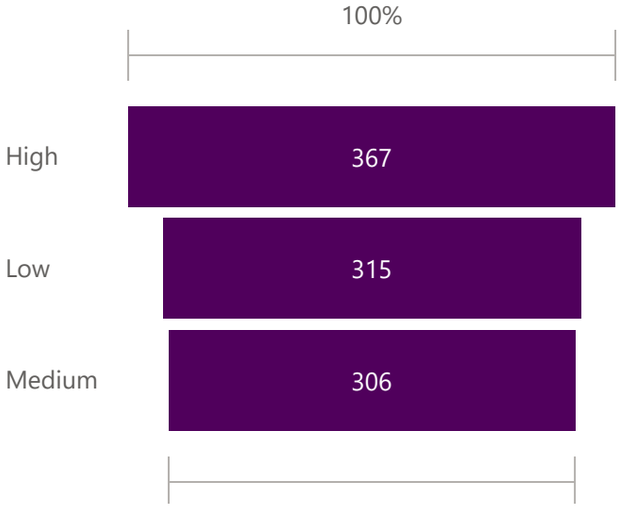
Sum of Distance_km and Sum of Shipping_Cost by Order_ID and Vehicle_Type



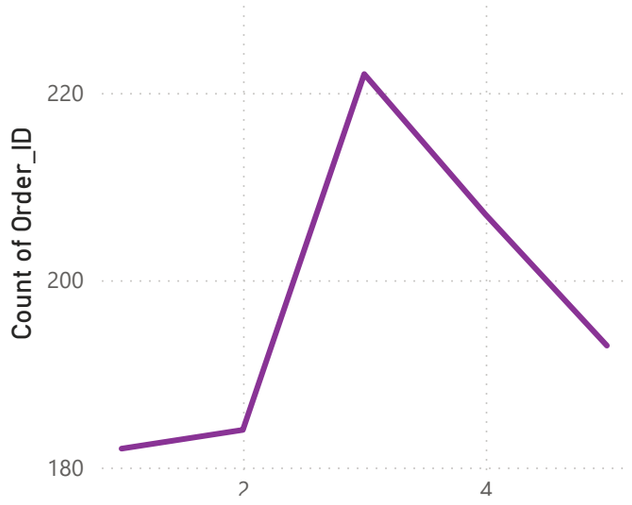
Sum of Shipping_Cost by Vehicle_Type



Count of Order_ID by Traffic

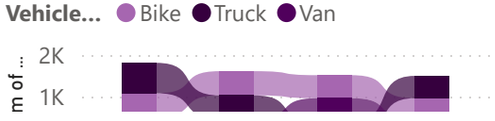


Count of Order_ID by Driver_Rating



Vehicle_Type	Clear	Cloudy	Fog	Rain	Tota
Bike	6	3	38	40	**8**
Truck	4	9	42	41	**9**
Van	4	4	38	26	**7**
Total	**14**	**16**	**118**	**107**	**25**

Sum of Delivery_Time by Weather and Vehicle_Type



AI Logistics Control Tower & Intelligent Operations Platform Operations Platform

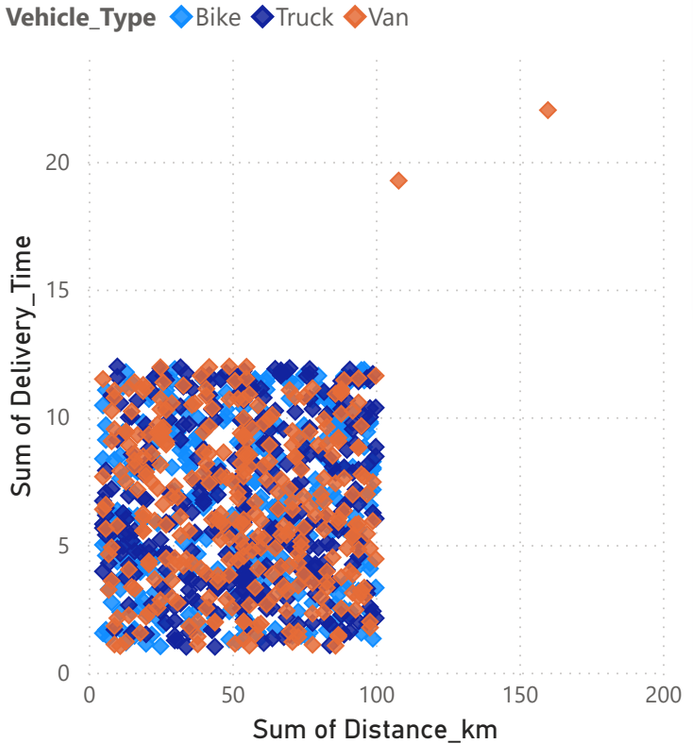
Delay percentage	Delay deliveries	On-Time Deliveries	Operational Status	Delay Risk Level
50.00%	494	494	High Delay	High
Delay %	Delayed Orders	On Time Orders	Operational Status	Delay Risk

Average of Shipping_Cost by Weather

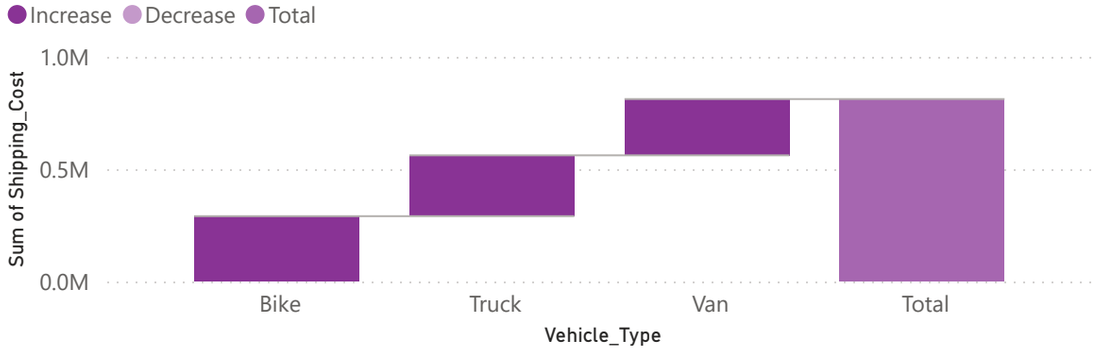


Traffic	Clear	Cloudy	Fog	Rain	**Total**
High	37	43	48	47	**175**
Low	38	47	51	36	**172**
Medium	41	28	40	38	**147**
Total	**116**	**118**	**139**	**121**	**494**

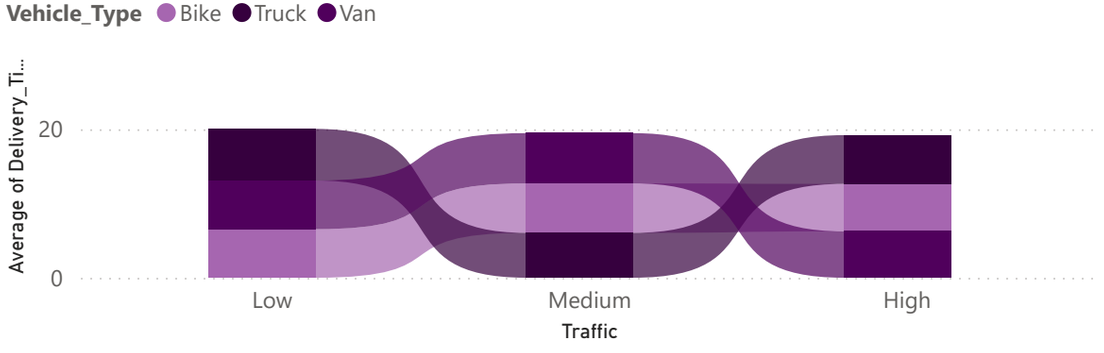
Sum of Distance_km and Sum of Delivery_Time by Order_ID and Vehicle_Type



Sum of Shipping_Cost by Vehicle_Type



Average of Delivery_Time by Traffic and Vehicle_Type



Business Recommendations

1.Route Optimization
The analysis indicates that optimizing delivery routes can significantly reduce travel time and minimize delivery delays. Efficient route planning helps improve on-time deliveries and reduces operational costs.

2.Driver Performance
Regular monitoring of driver ratings and performance can improve service quality. Providing training and performance incentives will help drivers maintain consistent and reliable delivery standards.

3.Traffic and Weather Management
Traffic congestion and weather conditions have a direct impact on delivery performance. Planning deliveries based on real-time traffic and weather information can reduce delays and improve scheduling efficiency.

4.Fleet and Cost Management
Selecting the appropriate vehicle for each delivery based on distance and shipment requirements helps optimize fleet utilization. Efficient fleet management also reduces fuel consumption and shipping costs.

5.AI-Based Delay Prediction
The machine learning model can identify deliveries that are likely to be delayed before dispatch. This enables logistics managers to take preventive actions and improve delivery reliability.

6.Business Intelligence
The Power BI dashboard provides real-time monitoring of key performance indicators such as total orders, delivery time, shipping cost, driver rating, and delayed orders. These insights support faster and more accurate business decisions.

7.Expected Outcome
Implementing these recommendations will reduce delivery delays, improve operational efficiency, optimize resource utilization, enhance customer satisfaction, and support data-driven decision-making. The Smart Logistics Intelligence Platform provides a scalable solution for improving logistics performance through Machine Learning and Business Intelligence.

Business Recommendations

8. AI Fleet Performance Monitoring
The AI-powered dashboard continuously monitors fleet performance using driver ratings, delivery time, and vehicle utilization. This enables managers to identify underperforming vehicles and improve operational efficiency.

9. Real-Time KPI Monitoring
The Power BI dashboard provides live monitoring of key logistics KPIs such as Delay Percentage, On-Time Deliveries, Fleet Performance Score, Average ETA, and Operational Status, supporting faster decision-making.

10. ETA Monitoring
The platform estimates the average delivery time (ETA) using historical logistics data. This helps improve customer communication and delivery planning.

11. Delay Risk Monitoring
The AI system identifies delay risk levels based on delivery performance indicators. High-risk deliveries can be monitored proactively to minimize service disruptions.

12. Smart Decision Support
By combining Machine Learning insights with interactive Power BI visualizations, the platform helps logistics managers optimize routes, reduce operational costs, improve customer satisfaction, and enhance overall supply chain performance.